

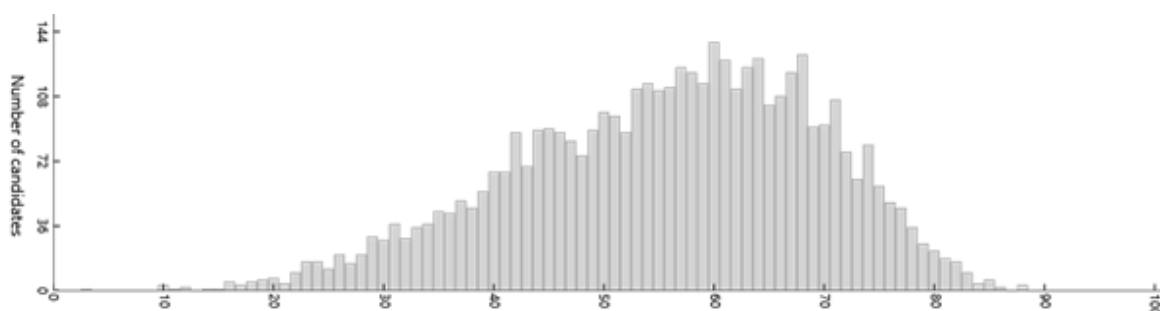


2020 ATAR course examination report: Human Biology

Year	Number who sat	Number of absentees
2020	4380	64
2019	4475	50
2018	4962	54
2017	4861	63

The number of candidates sitting and the number attempting each section of the examination can differ as a result of non-attempts across sections of the examination.

Examination score distribution–Written



Summary

Attempted by 4379 candidates Mean 55.61% Max 88.32% Min 0.00%

The paper discriminated well, producing scores from 0.00% to 88.32%. The paper was an appropriate length with most candidates attempting the final questions in the paper.

Section means were:

Section One: Multiple-choice	Mean 77.63%		
Attempted by 4379 candidates	Mean 22.09(/30)	Max 30.00	Min 0.00
Section Two: Short answer	Mean 48.66%		
Attempted by 4374 candidates	Mean 24.33(/50)	Max 41.82	Min 0.00
Section Three: Extended answer Unit 3	Mean 49.00%		
Attempted by 4303 candidates	Mean 4.90(/10)	Max 10.00	Min 0.00
Section Three: Extended answer Unit 4	Mean 42.87%		
Attempted by 4258 candidates	Mean 4.29(/10)	Max 10.00	Min 0.00

General comments

There was an increased proportion of candidates compared to previous years who did not attempt all question parts throughout the paper. Generally, those candidates who did attempt all questions in the paper were able to demonstrate good factual recall and understanding of the main concepts of the syllabus. In both Sections Two and Three, candidates could recall basic facts, but many often struggled to answer the in-depth analysis questions. Harder questions that were worth substantial marks had a notable proportion of candidates who did not attempt them.

Advice for candidates

- The examination is based on the syllabus and not a textbook. Candidates should expect a comprehensive coverage of most syllabus points in the examination; however, not every syllabus point will be examined. Although the same syllabus points may appear in consecutive examinations, questions will be structured in new and different contexts.

- Candidates must read questions carefully and ensure they understand the meaning of verbs used in the question. Candidates need to know the difference between a question requiring them to 'name' or 'identify' compared to one requiring them to 'describe' or 'explain'.
- Candidates are encouraged to try and answer every question. Candidates should always put something down as an answer, even if it is an educated guess. A non-attempt response only ensures a zero mark.
- Candidates need to focus on interpreting the question and providing a concise answer, rather than simply writing down all they have learnt on the topic. Stating key terminology or memorised facts will often not answer the entire question. Instead, candidates need to engage higher-order thinking skills and apply their knowledge.
- Candidates are reminded that no marks are available for restating information in their response from the question. Read the questions thoroughly and ensure the response covers the required question.
- The most thorough and complete responses in the extended section of the paper were produced by candidates who had completed a plan. Take time to plan your response and ensure all parts of the question are addressed.
- Candidates are encouraged to present annotated diagrams, charts or tables to construct responses to questions in the short and extended sections. This technique is particularly important in the extended answer section to help candidates write clear and precise answers and ensure that markers can easily follow and award marks for responses.

Advice for teachers

- Students require a detailed knowledge of all syllabus points and the ability to apply these points to new and different contexts. Simple recall and rote learning of facts are not enough for candidates to gain full marks in the examination. Candidates must be able to think critically and apply their knowledge to unique scenarios.
- There were several specific syllabus points that were not answered well in this examination. These include:
 - o graphing and mathematical skills
 - o treatment of Parkinson's disease with cell replacement therapy
 - o specificity of the immunity provided by vaccinations
 - o causes of the changes to allele frequencies in gene pools
 - o alpha and beta thalassemia
 - o differences between hominin skulls
 - o trends in tool culture.
- Teachers are reminded that the textbook is not the syllabus. Supplementary teaching is necessary and learning with other references and resource materials beyond the textbook is essential.
- Examination technique should be modelled and reinforced throughout the year during classroom lessons and assessments.
- Mathematical skills are a required element in the syllabus document and the list of these skills goes beyond basic mathematical and graphing skills. Give candidates ample practise at applying all required mathematical skills in a scientific method context.
- While the question on animal ethics was actually quite successfully answered in the paper, the question reflected the need to address the whole syllabus document.
- Students need to be given ample practise at decoding extended response questions as part of their lessons and school-based assessment program. They need substantial practise in breaking down a question and identifying the question components.

Comments on specific sections and questions

In Section One, mean scores ranged from 17% to 93%. Some questions were answered very successfully in this section and demonstrated outstanding factual recall of specific syllabus points such as PCR, diabetes, thermoregulation, studies of comparative anatomy

and the divisions of the nervous system. Section Two had mean scores ranging from 30.4% to 74.8%. Questions relating to homeostasis were completed successfully. The extended answer section was the most demanding section of the paper with the lowest overall mean of the three sections. The mean scores of the Unit 4 questions were comparable at 43.9% for Question 41 and 41.4% for Question 42. However, the mean scores of the Unit 3 questions varied greatly with a mean of only 30.6% for Question 39 and 55.9% for Question 40. The syllabus content answered most successfully in the extended answer section related to the endocrine system and biotechnology.

Section One: Multiple-choice (30 Marks)

This section had an overall mean of 77.63%. The maximum mark achieved in this section was 100% and the minimum mark was 0%. Generally, questions were well answered with only Questions 16, 19, 20, 21 and 29 having mean scores at or below 60%. Questions 5, 8, 12, 13, 18 and 26 were the easiest with mean scores all above 85%. Candidates performed well on straightforward recall style questions and found the more complex, multi-step questions challenging.

The data presented in Question 16 is a frequency score in a continuous set and as such is best plotted as a histogram. Many candidates appear to have confused column graphs and histograms. Question 19 had a low mean, with many candidates unable to distinguish the diagram as being representative of a sensory neuron. Question 20 proved difficult with many candidates not knowing why an antibiotic cannot work on a virus. Question 21 required basic mathematical skills to complete with many candidates struggling with the simple calculations. Question 29 reported a very low mean score with most candidates not being able to distinguish between the *Australopithecus afarensis* and *Australopithecus africanus* skulls. Many candidates selected option (c) identifying skull Y as *Paranthropus robustus* which was incorrect.

Question 1 attempted by 4379 candidates	Mean 0.73(/1)	Max 1	Min 0
Question 2 attempted by 4379 candidates	Mean 0.83(/1)	Max 1	Min 0
Question 3 attempted by 4379 candidates	Mean 0.76(/1)	Max 1	Min 0
Question 4 attempted by 4379 candidates	Mean 0.66(/1)	Max 1	Min 0
Question 5 attempted by 4379 candidates	Mean 0.92(/1)	Max 1	Min 0
Question 6 attempted by 4379 candidates	Mean 0.83(/1)	Max 1	Min 0
Question 7 attempted by 4379 candidates	Mean 0.76(/1)	Max 1	Min 0
Question 8 attempted by 4379 candidates	Mean 0.88(/1)	Max 1	Min 0
Question 9 attempted by 4379 candidates	Mean 0.86(/1)	Max 1	Min 0
Question 10 attempted by 4379 candidates	Mean 0.76(/1)	Max 1	Min 0
Question 11 attempted by 4379 candidates	Mean 0.76(/1)	Max 1	Min 0
Question 12 attempted by 4379 candidates	Mean 0.93(/1)	Max 1	Min 0
Question 13 attempted by 4379 candidates	Mean 0.88(/1)	Max 1	Min 0
Question 14 attempted by 4379 candidates	Mean 0.82(/1)	Max 1	Min 0
Question 15 attempted by 4379 candidates	Mean 0.77(/1)	Max 1	Min 0
Question 16 attempted by 4379 candidates	Mean 0.41(/1)	Max 1	Min 0
Question 17 attempted by 4379 candidates	Mean 0.84(/1)	Max 1	Min 0
Question 18 attempted by 4379 candidates	Mean 0.92(/1)	Max 1	Min 0
Question 19 attempted by 4379 candidates	Mean 0.60(/1)	Max 1	Min 0
Question 20 attempted by 4379 candidates	Mean 0.44(/1)	Max 1	Min 0
Question 21 attempted by 4379 candidates	Mean 0.59(/1)	Max 1	Min 0
Question 22 attempted by 4379 candidates	Mean 0.77(/1)	Max 1	Min 0
Question 23 attempted by 4379 candidates	Mean 0.75(/1)	Max 1	Min 0
Question 24 attempted by 4379 candidates	Mean 0.77(/1)	Max 1	Min 0
Question 25 attempted by 4379 candidates	Mean 0.74(/1)	Max 1	Min 0
Question 26 attempted by 4379 candidates	Mean 0.87(/1)	Max 1	Min 0
Question 27 attempted by 4379 candidates	Mean 0.61(/1)	Max 1	Min 0

Question 28 attempted by 4379 candidates	Mean 0.61(/1)	Max 1	Min 0
Question 29 attempted by 4379 candidates	Mean 0.17(/1)	Max 1	Min 0
Question 30 attempted by 4379 candidates	Mean 0.84(/1)	Max 1	Min 0

Section Two: Short answer (110 Marks)

This section had an overall mean of 48.66%. The maximum mark achieved on the section was 83.64% and the minimum mark was 0%. Questions 31, 33 and 34 all had mean scores above 55%. Question 31 focused on homeostatic feedback loops and was the most successfully completed question in this section. Questions 32, 35, 36, 37 and 38 had mean scores below 50%. Question 36 on scientific method and thalassemia was the most problematic question of section two. Generally, candidates were able to demonstrate a good recall of basic facts and knowledge. The higher order and application questions were successful in differentiating candidates and provided an opportunity for candidates to demonstrate their understanding of course content.

Question 31 attempted by 4365 candidates Mean 7.48(/10) Max 10 Min 0
This question was the most successfully completed in this section. All parts of the question had good mean scores and candidates demonstrated a very sound understanding of homeostatic feedback loops and the differences between the nervous and endocrine system. In part (b)(ii) candidates needed to describe the role of a receptor in a feedback loop. Candidates needed to state that a receptor detects the change, not simply receive the stimuli and secondly state that the receptor passes that information onto a modulator.

Question 32 attempted by 4363 candidates Mean 5.93(/12) Max 12 Min 0
Generally, the later parts of the question on neurotransmitters was answered well. However, the parts of the question on disease presented problems. Many candidates struggled to describe the structural characteristics of a bacterium. It is important for candidates to be able to distinguish between the features of bacterial and viral pathogens in order to understand how these pathogens cause disease, how they are transmitted and why antibiotics and antivirals are or are not effective in treating diseases caused by them. Part (b) on disease transmission was also answered poorly. Candidates did appear to understand the content, but could not construct an answer to fully describe the mode of transmission. Candidates needed to describe the whole transmission process, including how the pathogen can be released and how the pathogen can be introduced into a new host. Many candidates addressed only part of this process.

Question 33 attempted by 4368 candidates Mean 9.43(/16) Max 16 Min 0
Candidates were able to demonstrate a sound understanding of homeostasis of body fluids in this question. Generally, the graph in part (a) was completed to a high standard, but candidates often struggled to provide a valid title for their graph. Using terms like 'vs' and not adequately describing both the independent and dependent variables were common errors. In part (d)(ii) of the question, those candidates who named and described the role of all the components of the feedback loop clearly were able to get maximum marks.

Question 34 attempted by 4367 candidates Mean 8.47(/15) Max 15 Min 0
Candidates were able to demonstrate a sound understanding of the parts of the brain and protection of the brain in part (a). Part (b) relating to Parkinson's disease was completed poorly. A majority of candidates could not describe the cause of Parkinson's disease as being the degeneration of nerve cells of the brain and decreased dopamine production. This basic knowledge is essential to understanding how cell replacement therapy can be used to treat Parkinson's disease. Part (b)(ii) required candidates not to simply describe cell replacement therapy, but apply it to the treatment of Parkinson's disease. In the final part of the question, candidates could show they knew the structural features and pathways of the arc, but their answers were often worded poorly, lacked explicit explanation and omitted key terminology.

Question 35 attempted by 4327 candidates Mean 5.01(/12) Max 11 Min 0

In part (a) of the question the understanding of the causes of changes to allele frequencies in gene pools that was demonstrated by the candidates was substandard. The question used familiar wording, yet many candidates did not appear to comprehend the question. Candidates failed to identify that the stimulus information showed that gene flow would be restricted due to the geographical barrier (ocean) and that the different islands would have differing environmental selection pressures that in both cases lead to independent changes in allele frequencies across the islands of the region. This was intended to be a more difficult question, and the quality of answers was very poor. The latter parts of the question on radiocarbon and relative dating were answered well.

Question 36 attempted by 4345 candidates Mean 4.87(/16) Max 13 Min 0

This question was the least successfully completed in this section. It focused on scientific methodology and thalassemia, with a large number of candidates providing mediocre responses. In regard to scientific method, candidates often struggled to provide an adequate hypothesis. Many candidates failed to construct a statement that incorporated the independent and dependent variable. When analysing the experimental design, many candidates could not explain clearly the purpose of multiple control groups in an experiment that had multiple drug delivery methods or that data would need to be collected before and after testing to ensure the hypothesis could be assessed. Only a small proportion of candidates were able to describe the difference between the two thalassemia diseases. Part (f) was worded in a very similar way to the syllabus, but the vast majority of candidates could not successfully answer the question. It also had a very high rate of non-attempts. Often, candidates that did attempt the question were highly focused on the interaction between malaria and thalassemia, but missed the main difference between these two diseases that of alpha thalassemia being fatal in the homozygote while beta thalassemia is not.

Question 37 attempted by 4348 candidates Mean 4.75(/12) Max 12 Min 0

Candidates were able to demonstrate sound recall of passive and active immunity. However, other parts of the question presented problems. In part (a)(ii), responses from candidates often lacked precision and did not incorporate the important role of B cells and memory cells in how a vaccine provides immunity. In the final parts of the question it appeared that candidates did not read the question clearly. In part (b)(ii), candidates often described antibodies being passed in breast milk when the question asked how antibodies would be already present in a baby at birth. In part (c), many candidates seemed to miss that the child had contracted measles before their first birthday. As such, many candidates talked about immunity coming from the vaccine or from the mother's antibodies instead of the child's own immune response to the infection. Also, in this final question there was a lot of confusion shown around the MMR vaccine. A large proportion of candidates could not articulate clearly that the child who had already had measles would only have memory cells for the specific measles antigen and still needs the MMR vaccine to develop the specific memory cells for mumps and rubella.

Question 38 attempted by 4364 candidates Mean 7.59(/17) Max 16 Min 0

Many candidates were able to interpret a detailed phylogenetic tree as required in the first part of the question, but few candidates were able to transfer that skill to drawing their own interpretation of the data into a new tree diagram as required in part (c). The part of the question relating to how DNA sequencing can be used to construct a phylogenetic tree was completed poorly. Many candidates described the process of DNA sequencing instead of answering how the evidence is used. The final parts of the question on tool culture was completed poorly. Even the question on ordering the tools from youngest to oldest was done poorly, suggesting candidates struggled to identify the key trends in tool manufacture. Candidates appeared to lack the ability to use terminology correctly when discussing these concepts and often attempt to answer the questions without specificity.

Section Three: Extended answer Unit 3 (20 Marks)

This section had an overall mean of 49.00%. The maximum mark achieved on the section was 100% and the minimum mark was 0%. Question 40 was the most successfully answered question with a mean of 55.9%, while Question 39 achieved a mean of 30.6%. Generally, candidates who attempted questions using annotated diagrams and tables were able to construct more precise and understandable responses.

Question 39 attempted by 1115 candidates Mean 6.11(/20) Max 17.5 Min 0
This question was both the least attempted question and recorded the lowest overall mean score. All parts of the question were done poorly. In part (a), candidates often presented responses that lacked detail, failing to mention macrophages, pyrogens and interleukins in the fever response. In parts (b) and (c), many candidates did not appear to have the depth of knowledge on how antibiotics can work in different ways on different bacterial infections, nor could they provide a logical, scientific explanation of why stopping an antibiotic course early would allow the remaining bacterial population to increase in size, making the host feel sick again.

Question 40 attempted by 3188 candidates Mean 11.18(/20) Max 20 Min 0
This question was the most successfully completed of all extended response questions and also the most popular choice among candidates. All parts of the question had mean scores above 50% and were done relatively well. Particularly well answered was part (a) on how the anterior pituitary and the hypothalamus work together to release human growth hormone.

Section Three: Extended answer Unit 4 (20 Marks)

This section had an overall mean of 42.87%. The maximum mark achieved on the section was 100% and the minimum mark was 0%. Questions 41 and 42 had comparable mean scores of 43.9% and 41.4% respectively. Generally, candidates who attempted to answer the questions using annotated diagrams and tables were able to construct more precise and understandable responses.

Question 41 attempted by 2469 candidates Mean 8.78(/20) Max 20 Min 0
This question was slightly more favoured than Question 42 and had only a slightly higher mean score. Part (a) of the question required candidates to apply their understanding of founder effect and natural selection to the hypothetical situation of colonising Mars. While many candidates could describe the process of natural selection, only the more capable candidates were able to differentiate between founder effect and natural selection and relate their answer to the context required. Generally, part (b) on DNA sequencing was answered well. Candidates were able to obtain full marks on the question if they described gel electrophoresis only, but many candidates described the full Sanger sequencing process correctly.

Question 42 attempted by 1789 candidates Mean 8.27(/20) Max 19.5 Min 0
Both parts of the question presented problems for candidates. In part (a), many candidates simply did not write enough for 15 marks. Candidates needed to demonstrate a complete understanding of mutations including gene versus chromosomal mutations, germline versus somatic mutations and the differences between mutations caused by DNA replication, cell division and mutagens. Responses needed to have both depth and breadth. Part (b) was completed very poorly. The question required candidates to state the hominin jaw features and also to apply the evolution of this structure to the natural selection process. The majority of candidates could not construct a logical answer to this question.